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Predicting Animal Space-Use with Continuous-Time Markov Chain Movement Models

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Abstract. Remote animal telemetry data is widely used in ecology to assess animal space-use and evaluate utilization distributions (UDs). However, traditional empirical UD (like kernel density or Brownian bridge estimators) can suffer from “start bias” if tag deployments are too short for an animal to fully explore its habitat, heavily biasing results toward the initial release location. Recently, predictive UD have been proposed in an attempt to mitigate this by finding the long-run stationary distribution of a stochastic process that uses habitat covariates to describe movement. This stationary distribution is not directly dependent on starting location because it uses habitat variables to predict space-use in the future based on short term movements. Predictive UD have been calculated from Langevin diffusions as well as general continuous-time Markov chain models (CTMC). While log-linear closed form UD calculations are possible for Langevin models with coefficients directly relating habitat to space-use, CTMC UD have only been calculated numerically without parameters relating directly habitat to use. Here we present a general class of CTMC models and show they have a closed form log-linear predictive UD. CTMCs have several benefits over Langevin models: (1) the exact likelihood can be used without approximation, (2) both categorical and continuous habitat variables can be used, and (3) fixed barriers to movement (e.g., land in marine settings) are automatically accounted for as part of the model. We developed the R package `walk` for fitting these models in practice and it is used to demonstrate explicit UD estimation from CTMC models with a data set of Hawaiian false killer whales (*Pseudorca crassidens*). We also illustrate a 2-stage meta-analysis method for population inference from individual UD.

Key words: Hidden Markov model, resource selection, Stationary distribution, Satellite telemetry, Utilization distribution

1 Introduction

Any quick literature search will reveal that remote collection and analysis of animal locations, or location telemetry data, have become a ubiquitous and important tool in the advancement of animal ecology. A perennial subject within animal telemetry analysis is the assessment of space-use. The topic of animal space-use is often used in conjunction with the terms home range (Burt 1943) or resource selection (Johnson 1980), but here we are specifically concerned with the concept of utilization distributions (UD; Powell and Mitchell 2012). UDs are a spatial probability distribution that describes an animal’s location at any random time. As opposed to standard (Keating and Cherry 2009) or correlated kernel density estimates such as Brownian bridge (Horne et al. 2007) or AKDE (autocorrelated kernel density estimator; Fleming et al. 2015; Calabrese et al. 2016) we consider UD models that explicitly relate habitat variables to use through the log-linear form,

$$\mathbf{u}(\mathbf{s}) \propto \exp\{\mathbf{x}(\mathbf{s})'\boldsymbol{\beta}\}, \quad (1)$$

where $\mathbf{u}(\mathbf{s})$ is the UD at location $\mathbf{s}$ and $\mathbf{x}(\mathbf{s})$ are habitat covariates at location $\mathbf{s}$. Inference for parametric spatial UDs from temporally correlated telemetry data can be challenging (Johnson et al. 2013). Estimates that ignore the temporal aspect can suffer from start-bias (Whitehead and Jonsen 2013; Calabrese et al. 2016) if the individual does not have a long enough deployment to fully explore its home range. Also, (Alston et al. 2023) illustrated treating locations as independent will underestimate uncertainty in $\boldsymbol{\beta}$ inference in resource selection models.

To account for autocorrelated telemetry data, predictive UDs have been proposed which are derived by fitting an autocorrelated movement model to the telemetry data, where the stationary distribution of the movement model is treated as the UD. Predictive UDs can reduce or eliminate start bias because they predict where the individual will be in

the long-run regardless of the starting location (Whitehead and Jonsen 2013). Secondly, the UD parameters are estimated by explicitly modeling movement with autocorrelation which avoids the predicament of underestimation of parameter uncertainty by treating the data as independent. Barnett and Moorcroft (2008) and Moorcroft et al. (1999) considered mechanistic stochastic differential equation (SDE) movement models but the resulting UD was time consuming to numerically assess. This same approach was also used to scale step-selection models to the resulting UD (Potts and Börger 2023). Langevin diffusions were recently proposed as an alternative SDE model because they have a log-linear form (1) stationary distribution (Michelot et al. 2019; Dupont et al. 2026). However, only continuously valued habitat variables can be used in these models. Traditionally, Langevin models could not directly account for spatial barriers, however, recent developments have allowed researchers to fit models that have a soft penalty that strongly discourages movement into prohibited areas (Dupont et al. 2026; Delporte et al. 2026).

Continuous-time Markov chain (CTMC) movement models have become another type of movement model that can easily incorporate habitat variables to describe movement. We provide a brief description here, but more technical details can be found in (Norris 1998). In a CTMC movement model the geographical study domain is partitioned into cells or other discrete locations indexed by $\mathcal{C} = \{1, \dots, n\}$ (See Figure 1B-E). Upon moving into a cell, an individual will remain for an exponentially distributed amount of time, then move to a neighboring cell. Each cell, i , has a set of neighboring cells $\mathcal{N}_i = \{j \in \mathcal{C} : i \sim j\}$, where $i \sim j$ means that an animal can go from cell i to cell j directly. For example, in a raster partition, neighboring cells might share a border (rook neighborhood) or a corner point (queen neighborhood). CTMC models are defined by the conditional movement rates between cells, $\lambda_{ij} \geq 0$. The total rate of movement away from cell i is $\Lambda_i = \sum_{j \in \mathcal{N}_i} \lambda_{ij}$. After an individual enters cell i it will reside there an exponential, mean Λ_i^{-1} , amount of time, then transition to a neighboring cell with probability $P_{ij} = \lambda_{ij}/\Lambda_i$. Two important

benefits of CTMC models over SDEs is that categorical habitat variables can be used and CTMCs can be formulated as large sparse hidden Markov models (HMMs) so that efficient evaluation of the exact likelihood is possible. To this point most analyses of satellite location telemetry data with CTMCs have also relied on imputation, for example, Hanks et al. (2015); Wilson et al. (2018); Johnson et al. (2021), but see Pedersen et al. (2011) for an archival tag CTMC HMM. One additional benefit of CTMCs relative to SDEs is that complex barriers can be easily accounted for simply by the definition of the cell partitions (Wilson et al. 2018), that is, for a cell j where the animal cannot travel, $\lambda_{ij} \equiv 0$ for all $i \in \mathcal{N}_j$. Or, in other words $j \notin \mathcal{C}$. If all the rates are placed in an $n \times n$ matrix $\mathbf{Q}$ with $(-\Lambda_1, \dots, -\Lambda_n)$ on the diagonal and λ_{ij} in the i th row, j th column then one can also represent a CTMC as the discrete-space SDE,

$$\frac{d}{dt}\mathbf{u}_t = \mathbf{Q}\mathbf{u}_t,$$

where $\mathbf{u}_t$ is the distribution of the individuals location at time t . The solution to this SDE is,

$$\mathbf{u}_t = \mathbf{\Gamma}(t)\mathbf{u}_0,$$

where $\mathbf{\Gamma}(t) = \exp(\mathbf{Q}t)$ is the transition probability matrix (TPM) and “exp” is the matrix exponential function. If the rates and study area are constructed such that: (1) rates are constant through time (homogeneity), (2) any cell can be reached from any other cell given sufficient time (irreducibility), and (3) once a cell is departed the probability of returning in a finite amount of time is > 0 (positive recurrence), then the CTMC will have a stationary distribution $\mathbf{u}_t \rightarrow \mathbf{u}$ as $t \rightarrow \infty$ (Whitehead and Jonsen 2013; Wilson et al. 2018). This stationary distribution is the eigenvector of $\mathbf{Q}$ associated with the zero eigenvalue. Therefore, a general CTMC movement model does not have the interpretable log-linear form (1) with parameters that directly relate use to habitat variables as it must be

95 numerically determined.

96 In order to blend the benefits of the CTMC based movement model with the
97 closed-form, log-linear UD of the Langevin SDE model we propose a class of CTMC
98 models for which one can obtain closed-form log-linear predictions of the UD from the
99 fitted movement model parameters. Markov Chain Monte Carlo (MCMC) theory is used to
100 verify that these specific models possess the desired form for their stationary distributions
101 (e.g., Michelot et al. 2018; Blackwell 2026) and we develop the HMM likelihood for fitting
102 the model, specifically for normally distributed location errors. To fit these models, we
103 developed the R package `walk*` (Johnson 2026) to estimate parameters of general CTMC
104 models using the exact HMM likelihood (McClintock et al. 2020). To our knowledge this
105 has not been formulated and used for CTMC-based animal movement using noisy satellite
106 location telemetry data. The process of fitting models and making UD inference is
107 demonstrated with telemetry data from a false killer whale (*Pseudorca crassidens*)
108 population in the Hawaiian Islands. In addition to individual UD inference we also adapt
109 the 2-stage meta-analysis method of Johnson et al. (2022) for approximating population
110 UD inference from a Bayesian random effects CTMC model.

111 2 Explicit Utilization Distribution CTMC Models

112 General CTMCs do not have closed-form stationary distributions, however, we consider a
113 class of CTMC useful for animal movement modeling for which we can directly evaluate
114 the stationary distribution. The movement rates of this class have the form,

$$\log(\lambda_{ij}) = \mathbf{x}'_{1i}\boldsymbol{\alpha} + \mathbf{x}'_{2j}\boldsymbol{\delta} + (\mathbf{x}_{3j} - \mathbf{x}_{3i})'\boldsymbol{\gamma} + \mathbf{x}'_{4ij}\boldsymbol{\eta}, \quad (2)$$

*Available from: <https://dsjohnson.r-universe.dev/walk>

115 where $\mathbf{x}_{1i}$ is a set of habitat covariates recorded in the current cell i , $\mathbf{x}_{2j}$ are other
 116 covariates recorded in cell $j \in \mathcal{N}_i$, $\mathbf{x}_{3j} - \mathbf{x}_{3i}$ are local differences of a third set of habitat
 117 covariates, and $\mathbf{x}_{4ij}$ are symmetric pair specific covariates ($\mathbf{x}_{4ij} = \mathbf{x}_{4ji}$) such as, say,
 118 distance between i and j . The covariates in $\mathbf{x}_1$ only control residence time and do not
 119 affect which cell an individual might choose upon moving. The variables in $\mathbf{x}_2$ primarily
 120 function to specify preference for neighboring cells based on the magnitude of the
 121 covariates in the neighboring cells, while the variables included via local differences, $\mathbf{x}_3$,
 122 control movement based on relative habitat differences, similar to habitat gradient effects
 123 in Langenvin diffusions. Now that we have a defined class of CTMC movement models we
 124 can evaluate the stationary utilization distribution that results from this CTMC diffusion.
 125 The symmetric covariates, $\mathbf{x}_{4ij}$, can be included to add effects such as distance between
 126 cells. For example, in a queen neighborhood structure the diagonal cells are slightly farther
 127 (by a factor of $\sqrt{2}$) from the resident cells than the non-diagonal neighbors.

128 **2.1 Evaluating the stationary utilization distribution**

129 To evaluate the stationary distribution and make sure one exists we first check the 3
 130 convergence criteria. First, all rates are homogeneous through time. Second, all movement
 131 rates in (2) are positive for finite covariates and coefficients, therefore if $i \sim j$, then $j \sim i$.
 132 Finally, if the study area is not divided into separate regions for which inter-region moves
 133 are impossible, then the positive rates imply irreducibility and positive recurrence.

134 We have established that a CTMC with rates defined by (2) will have a stationary
 135 utilization distribution. To assess what the actual form of the UD is, we begin with
 136 another theoretical result for evaluating a CTMC stationary distribution, that is,

$$u_i \propto \pi_i / \Lambda_i, \tag{3}$$

where $\boldsymbol{\pi}$ is the stationary distribution of the embedded Markov chain of the CTMC (Norris 1998; Wilson et al. 2018). The embedded Markov chain is the ordered list of cells visited by a CTMC realization, disregarding the actual times when it moves. The embedded Markov chain is a true discrete-time Markov chain with transition probabilities, $P_{ij} = \lambda_{ij}/\Lambda_i$. Therefore, we can work in the discrete-time to find the stationary distribution of the CTMC.

We can use Markov Chain Monte Carlo (MCMC) theoretical results to show that a CTMC movement model has a stationary distribution of log-linear form (e.g., Michelot et al. 2018; Blackwell 2026). To verify the closed-form stationary distribution we need only check that the MCMC reversibility condition is satisfied for a given $\boldsymbol{\pi}$, that is,

$$\pi_i P_{ij} = \pi_j P_{ji},$$

for all i and j in $\mathcal{C}$ (Blackwell 2026). By inserting P_{ij} and P_{ji} into the equation, one can check that it is satisfied for $\pi_i \propto \exp(-\mathbf{x}'_{1i}\boldsymbol{\alpha} + \mathbf{x}'_{2i}\boldsymbol{\delta} + 2\mathbf{x}'_{3i}\boldsymbol{\gamma})\Lambda_i$ (See online supplement S1). This implies, using (3), the stationary UD for the full CTMC model is,

$$\begin{aligned} u_i &\propto \exp(-\mathbf{x}'_{1i}\boldsymbol{\alpha} + \mathbf{x}'_{2i}\boldsymbol{\delta} + 2\mathbf{x}'_{3i}\boldsymbol{\gamma}) \\ &= \exp(\mathbf{x}'_i\boldsymbol{\beta}), \end{aligned} \tag{4}$$

where $\mathbf{x}_i = (\mathbf{x}'_{1i}, \mathbf{x}'_{2i}, \mathbf{x}'_{3i})'$ and $\boldsymbol{\beta} = (-\boldsymbol{\alpha}', \boldsymbol{\delta}', 2\boldsymbol{\gamma}')'$. Thus, the utilization distribution for the proposed class of CTMC movement models has the desired closed log-linear form (1).

2.2 Parameter Estimation

Previous estimation of CTMC movement parameters using satellite location telemetry data used an imputation method (see Scharf et al. 2017) which fills in missing cell locations and

transition times between location acquisition times (Hanks et al. 2015; Wilson et al. 2018; Johnson et al. 2021). Even though the imputation method attempts to propagate the uncertainty of the missing locations, it is unclear how this affects estimation of CTMC parameters. Therefore we propose using the exact, although potentially more computationally expensive, HMM formulation of the model.

2.2.1 Satellite telemetry data specification and processing

For the proposed analysis we consider satellite telemetry data for an individual of the form $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_K\}$, where $\mathbf{y}_k = (\mathbf{s}_k, t_k, \boldsymbol{\psi}_k)$, $\mathbf{s}_k$ is the observed location of the individual within the study area at time t_k and $\boldsymbol{\psi}_k$ represents information or parameters governing location uncertainty. For example, $\boldsymbol{\psi}_k$ might represent the major and minor error ellipse lengths of a Kalman Filtered Argos location (McClintock et al. 2015) or a known error radius for GPS locations, e.g., 20m. From these locations we translate the continuous-space data locations and uncertainty to the discrete space, $\mathcal{C}$, using a function, say, $p(\mathbf{y}_k, \mathcal{C}) = \mathbf{p}_k = (p_{1k}, \dots, p_{nk})$, where p_{ik} is the probability that the animal is located within cell i at time t_k . We used a general form because the function, p , will depend on how location uncertainty is assessed for the specific satellite and tag hardware. However, uncertainty in the true location, $\boldsymbol{\mu}_k$, is often represented by the bivariate Normal distribution $\boldsymbol{\mu}_k \sim N(\mathbf{s}_k, \boldsymbol{\Sigma}_k)$, where $\boldsymbol{\psi}_k = \boldsymbol{\Sigma}_k$. In that case, p_{ik} would represent the mass of the Normal distribution contained within the i th cell. For a raster grid partitioning of the study area this can be calculated by,

$$p_{ik} = \text{CDF}(x_{i2}, y_{i2}) - \text{CDF}(x_{i2}, y_{i1}) - \text{CDF}(x_{i1}, y_{i2}) + \text{CDF}(x_{i1}, y_{i1})$$

where $[x_{i1}, x_{i2}] \times [y_{i1}, y_{i2}]$ is the bounding box for the i th cell and CDF is the bivariate normal cumulative density function with mean $\mathbf{s}_k$ and covariance $\boldsymbol{\Sigma}_k$ (see Figure S1 for

illustration of $\mathbf{p}_k$). A Monte Carlo option for very general error distributions would be to draw a sufficiently large sample from the distribution and assess the proportion of samples that falls within each cell. If the error distribution is known, this only has to be done one time.

2.2.2 Likelihood calculation

For an individual, the likelihood is calculated with the matrix equation (Zucchini et al. 2016; McClintock et al. 2020),

$$[\mathbf{y}|\boldsymbol{\alpha}, \boldsymbol{\delta}, \boldsymbol{\gamma}, \boldsymbol{\eta}] = \mathbf{u}'_1 \mathbf{P}(\mathbf{y}_1) \boldsymbol{\Gamma}(t_2 - t_1) \mathbf{P}(\mathbf{y}_2) \cdots \boldsymbol{\Gamma}(t_K - t_{K-1}) \mathbf{P}(\mathbf{y}_K) \mathbf{1} \quad (5)$$

where $\mathbf{u}_1$ is the location probability distribution at t_1 , $\mathbf{P}(\mathbf{y}_k) = \text{diag}(\mathbf{p}_k)$, $\boldsymbol{\Gamma}(t_k - t_{k-1})$ is the TPM evaluated over the interval $t_k - t_{k-1}$, and $\mathbf{1}$ is a vector of 1s. The natural choice for $\mathbf{u}_1$ is $\mathbf{u}$, the stationary distribution defined by the CTMC movement model, but there are other choices. For example, if a telemetered individual was released at a known location after tagging that may not be representative of its usual movement, then $\mathbf{u}_1$ would be a vector of all 0s except for the cell containing the release site, which would be a 1. In the false killer whale example below, we use $\mathbf{u}_1 = \mathbf{u}$ because they were tagged where they were encountered in the wild. In the R package `walk` we use scaling methods to avoid numerical underflow as well as the efficient sparse matrix uniformization methods presented in Sherlock (2021) to calculate products of the form $\mathbf{v}' \exp\{\mathbf{Q}(t_k - t_{k-1})\}$.

In addition to the matrix exponential evaluation numerical savings, another optional approach to save computational effort is to set a censoring value for the maximum Λ_i value. The numerical methods of Sherlock (2021) require more matrix operations for higher values of $\max(\Lambda_i)$. For some cells, the log linear λ_{ij} model may produce rates of movement that are unrealistically high in some parts of the study area. With respect to UDs this

would imply that an individual will not remain in that cell and the $u_i \approx 0$. Hewitt et al. (2023) considered a logit rate model that restricts this behavior, but that model does not allow us to construct a CTMC with an explicit stationary distribution. So, we propose picking a censoring value, say, Λ^* and setting $\lambda_{ij}^* = (\Lambda^*/\Lambda_i)\lambda_{ij}$ if $\Lambda_i > \Lambda^*$. If Λ^* is sufficiently large, this will have negligible impact on $\mathbf{u}$ estimation but can dramatically help likelihood computation. One reasonable way to choose Λ^* is to select the minimum amount of time that an animal would likely reside in a cell, say r^* , then set $\Lambda^* = 1/r^*$. This implies that the expected residency time $\geq r^*$ for all cells. The effect on $\mathbf{u}$ can be verified by comparing the exact, numerically calculated $\mathbf{u}$ from the censored rate model (Wilson et al. 2018) to $u_i \propto \exp(\mathbf{x}_i'\boldsymbol{\beta})$. If discrepancies are sufficiently large, Λ^* can be increased until the discrepancy is within tolerance. In the resulting model, if $\max(\Lambda_i) \leq \Lambda^*$ the UD's are identical.

Once the CTMC movement model has been fitted, the maximum likelihood estimate, $\hat{\boldsymbol{\beta}}_{\text{CTMC}} = (\hat{\boldsymbol{\alpha}}', \hat{\boldsymbol{\delta}}', \hat{\boldsymbol{\gamma}}')'$, needs to be transformed to the UD coefficients via $\hat{\boldsymbol{\beta}} = \mathbf{M}\hat{\boldsymbol{\beta}}_{\text{CTMC}}$ and $\mathbf{M} = \text{diag}((-1'_{\alpha}, 1'_{\delta}, 21'_{\gamma})')$. Thus, the large sample covariance matrix of the UD coefficients is $\hat{\mathbf{S}} = \mathbf{M}\hat{\mathbf{S}}_{\text{CTMC}}\mathbf{M}$, where $\hat{\mathbf{S}}_{\text{CTMC}}$ is the Hessian-derived large sample covariance matrix of $\hat{\boldsymbol{\beta}}_{\text{CTMC}}$. Then the UD estimate is $\hat{u}_i = \exp(\mathbf{x}_i'\hat{\boldsymbol{\beta}}) / \sum_{j=1}^n \exp(\mathbf{x}_j'\hat{\boldsymbol{\beta}})$. If only the standard errors are desired, one can simply multiply the CTMC coefficient standard errors by the multipliers, but if the entire covariance matrix is desired matrix multiplication is necessary. Variance of the fitted UD can be estimated via a simple Monte Carlo approach of drawing iterations, $\boldsymbol{\beta}^{(r)} \sim N(\hat{\boldsymbol{\beta}}, \hat{\mathbf{S}})$ then calculating $\mathbf{u}^{(r)}$. After several draws $\text{Var}(\hat{\mathbf{u}})$ can be estimated as the sample variance of the $\mathbf{u}^{(r)}$.

2.2.3 Population-level inference

Individual estimates of UD coefficients and predicted UD's can be difficult to interpret for population inference and management considerations. Therefore, assuming a sample of

representative individuals from the population, we suggest the more appropriate model to fit is the Bayesian random effects model,

$$[\mathbf{y}_1, \dots, \mathbf{y}_N | \boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_N] = \prod_{l=1}^N [\mathbf{y}_l | \mathbf{M}^{-1} \boldsymbol{\beta}_l, \boldsymbol{\eta}_l]$$

$$[\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_N | \boldsymbol{\theta}, \boldsymbol{\Sigma}] = \prod_{l=1}^N N(\boldsymbol{\beta}_l | \boldsymbol{\theta}, \boldsymbol{\Sigma})$$

where $\mathbf{y}_l$ is the observed data for individual $l = 1, \dots, N$, $\mathbf{M}^{-1} \boldsymbol{\beta}_l = (\boldsymbol{\alpha}'_l, \boldsymbol{\delta}'_l, \boldsymbol{\gamma}'_l)'$ and $\boldsymbol{\eta}_l$ are the movement parameters for individual l , and the prior distribution for the random population mean coefficient and variance is $[\boldsymbol{\theta}][\boldsymbol{\Sigma}]$. Even if individual inference is desired, it is useful to fit this hierarchical model because it allows us to share information between individuals via the posterior distribution $[\boldsymbol{\beta}_l | \mathbf{y}_1, \dots, \mathbf{y}_N]$. If we assume that there is some similarity in movement between individuals then we can leverage all data for better individual inference. Even though the random effects model would be beneficial, however, due to the computational demands for fitting the CTMC model to one individual, a full joint random effects model is not currently feasible.

Although we cannot fit the full random effects model all at once, we can tailor the general 2-stage meta-analysis methods of Johnson et al. (2022) to approximate inference for the hierarchical CTMC model using maximum likelihood estimates for each individual. In the first stage individual MLEs, $\hat{\boldsymbol{\beta}}_l$ and covariance matrices, $\hat{\mathbf{S}}_l$ are obtained using (5). In the second stage the MLEs are used as responses in the multivariate linear mixed model,

$$\hat{\boldsymbol{\beta}}_l = \boldsymbol{\theta} + \mathbf{z}_l + \boldsymbol{\epsilon}_l; \quad l = 1, \dots, N$$

where the random effect is $[\mathbf{z}_l | \boldsymbol{\Sigma}] = N(\mathbf{0}, \boldsymbol{\Sigma})$ and the error term is $[\boldsymbol{\epsilon}_l | \hat{\mathbf{S}}_l] = N(\mathbf{0}, \hat{\mathbf{S}}_l)$. REML or Laplace approximation procedures are used to fit this model subject to the prior constraint of $[\boldsymbol{\theta}][\boldsymbol{\Sigma}]$. Following second stage model fitting we can extract the approximate maximum *a posteriori* (MAP) estimates $\bar{\boldsymbol{\theta}}$ and $\bar{\boldsymbol{\beta}}_l = \bar{\boldsymbol{\theta}} + \bar{\mathbf{z}}_l$, where $\bar{\boldsymbol{\beta}}_l$ is the approximate

MAP of the posterior $[\beta_l | \mathbf{y}_1, \dots, \mathbf{y}_N]$.

3 Illustration: False Killer Whale Space-Use

In this section, we will apply the modeling approach described in the previous sections to a telemetry dataset from a population of endangered false killer whales (FKW) in the Hawaiian Islands (Baird 2016). This particular population is resident to the insular waters of the main Hawaiian Islands (MHI; includes Kaua‘i/Ni‘ihau, O‘ahu, Maui Nui, and Hawai‘i Island) and is comprised of four social clusters, or groups of close family members and regular associates (Baird et al. 2008; Baird 2016; Martien et al. 2019; Mahaffy et al. 2023). These whales are known to move quickly and frequently among island areas throughout their range (Baird et al. 2010, 2012; Baird 2016). Due to inherent challenges associated with tracking apex marine predators, this dataset is characterized by variable deployment durations and irregular location data streams. Considering these factors may drive start bias, as well as their expansive range, we deem this an appropriate dataset to empirically demonstrate the utility of predictive UD. For the purposes of this assessment, we only focus on movements of 5 individuals that were tagged in 2020 and 2021. There is at least one representative from each social cluster in the data. Table S1 (online supplement) provides a summary of the five individuals. Because this analysis is intended to illustrate the modeling framework using a small subset of available telemetry data, the resulting UD should not be interpreted as a comprehensive estimate of population-level space use or stock range.

3.1 Satellite tag data

False killer whale satellite tag deployment methods have been described in detail elsewhere (see Baird et al. 2010, 2012) so they are only briefly summarized here. Individuals were

tagged with Argos transmitting SPOT6 ($n = 4$) and SPLASH10-F ($n = 1$) tags during dedicated small boat survey efforts throughout the main Hawaiian Islands (Baird et al. 2012, 2021; Baird 2016; see Baird et al. 2013 for details on surveys). Deployment duration for these data ranged from 12 days to 81 days (median=40 days). Argos location data were processed through the Douglas-Argos Filter (accessed via Movebank; Kranstauber et al. 2011; Douglas et al. 2012) to remove erroneous locations based on unrealistic travel speeds and turning angles. The final analytical dataset consisted of 5 trajectories and 4,009 filtered Argos and Fastloc[®]-GPS (SPLASH10-F tag) locations (Figure 1A). For the Argos locations, we used the Kalman-filter diagnostic ellipse data to construct error covariance matrices (McClintock et al. 2015). For the Fastloc[®]-GPS locations, we used the 95% quantile error ranges from Dujon et al. (2014), based on the number of satellites acquired, as twice the standard deviation of the normal error.

3.2 Study area and habitat variables

The MHI FKW stock boundary described in Bradford et al. (2015) was used as the study-area boundary for this illustrative UD analysis (Figure 1A). We used two modeled global oceanographic variables from E.U. Copernicus Marine Service: (1) Sea Surface Temperature (SST; °C; DOI: <https://doi.org/10.48670/moi-00016>) and (2) Sea Level Anomaly (SLA; m; DOI: <https://doi.org/10.48670/moi-00148>) to model movement and space-use. SST is available at 0.082° resolution and SLA is available at 0.125° degree resolution. In order to match resolution we chose to project the SLA raster to the resolution of the SST raster to avoid losing oceanographic information. Because we are using these variables to predict long term space-use, we used a single layer for each variable that was the 2020-2021 daily average in the study area. This allowed us to model movement with respect to typical oceanographic conditions (Figure 1B-C).

In addition to the oceanographic variables, we used distance from land (m) to model FKW’s resident nature in the nearshore MHI. Also, we constructed a depth residual variable by using data from the global, 30 arc-second ETOPO elevation model (NOAA National Centers for Environmental Information 2022) and removing the linear trend of distance from land through simple linear regression. The distance and depth variables were projected to the 0.082° SST resolution as well. Then all data, including telemetry data, were projected to the Hawaii Albers Equal Area projection, ESRI:102007 (Figure 1D-E)

3.3 CTMC model and hierarchical inference

Notationally, the CTMC movement rate model we used to model the movement of each individual $l = 1, \dots, 5$ was

$$\log \lambda_{ij} = \underbrace{(\text{Intercept}) + \text{DEPTH}_i + \text{SST}_i + \text{SST}_i^2 + \text{SLA}_i + \text{SLA}_i^2}_{\mathbf{x}_1 \text{ portion}} + \underbrace{[\text{DIST}_j - \text{DIST}_i]}_{\mathbf{x}_3 \text{ portion}}$$

In this model differences of the distance from land variable were used as a movement kernel between cells i and j . This allowed individuals to prefer movement towards (or away from) land at the same rate no matter how far away they are. In addition SST, SLA, and DIST were Z -transformed for numerical stability of the HMM likelihood. The DEPTH residual was already on a nearly standard normal scale and DIST was only used as a difference, so values were not large enough to cause numerical issues.

In the first stage the CTMC model was fitted separately to the data from each individual using normal location errors and the HMM likelihood (5). To determine the censor value for the total movement rate we selected the minimum expected residency time to be $r^* = 1/3$ minutes. This implies a maximum total movement rate of $\Lambda^* = 60/(1/3) = 180$ transitions per hour. Here, the $r^* = 1/3$ was chosen because it represented a small amount of time such that expected residency in a cell for a shorter time

would imply negligible utilization. This value would likely be different for different species.

The `nimble` R package (de Valpine et al. 2017) was used to optimize the second stage multivariate linear mixed model using Laplace approximation (de Valpine et al. 2017). The random effects and prior distributions used for the CTMC/UD coefficients are,

$$\begin{aligned} [\theta_w] &= N(0, \tau_w^2); \quad [\tau_w] = \text{Exponential}(1) \\ \Sigma &= \text{diag}(\sigma_1^2, \dots, \sigma_7^2); \quad [\sigma_w] = \text{Exponential}(1) \end{aligned}$$

for $w = 1, \dots, 7$. The exponential hyperpriors for the variance parameters were added to penalize the complexity of the model to perform selection for the population level parameters (Hooten and Hobbs 2015; Simpson et al. 2017).

3.4 Results

Individual results were generally consistent, although individual PcTag074 did not have any significant effect of SLA on its UD and PcTag070 had a negative effect of the linear SST term (Figure 2). All other coefficients were similar in direction, i.e., less than or greater than 0. This bolsters the idea of using the 2 stage meta-analytic approach to share data to improve individual estimates. The second stage individual coefficient estimates, $\bar{\beta}_l$ illustrate that there is only a significant amount of individual variation in the DEPTH and SLA² coefficients (Figure 2). All other second stage coefficients are shrunk to the population mean estimates, $\bar{\theta}$ with nearly identical approximate credible intervals (Figure 2). There was only 1 nonsignificant population-level coefficient, the linear SLA term. Correspondingly, the penalized model complexity priors shrunk the SLA term to zero. All other population coefficients were significantly different from zero. The DEPTH and DIST coefficients were negative as expected because this population inhabits shallower waters close to the MHI without venturing too far away. The zero SLA coefficient combined with

the negative SLA^2 term implies a maximum effect at 0, but we used A Z -transform, therefore, the maximum effect occurs at the mean original SLA value. Both SST coefficients are positive implying that the typical individual is probably located in the warmest SST conditions in the study area.

Unsurprisingly, the similarity in coefficient estimates produced similar predicted UD at the first stage (Figure S1, online supplement). Individual PcTag074 had a more notably diffuse UD, the other four individuals had very similar looking UD. This may be due to the lack of SLA effect on the UD of PcTag074. After the second stage, the UD are even more similar as the second stage coefficients are essentially the same at the population mean coefficients (Figure 3). The UD created from the population mean coefficients is illustrated in Figure 3 (Typical). This is the UD of a so called “Typical” or median individual, not the population average UD because of the log link between the coefficients and the UD. Even after the second stage the coefficients for PcTag074 lack any SLA effect, so the UD is the only different one from the typical UD.

In order to check that the rate truncation did not adversely affect the estimated UD we compared the truncated numerical Eigen decomposition UD ($u_{ei,i}$; Wilson et al. 2018), which is the exact UD for the censored movement models that was used in the first stage, with the exponential closed form version using Bhattacharyya’s Affinity (BA) (Bhattacharyya 1943),

$$BA = \sum_{i \in \mathcal{C}} \sqrt{u_{ei,i} \cdot u_i}.$$

The values of BA range from 0 (no similarity) to 1 (identical distributions) (Fieberg and Kochanny 2005). The proportion of study area cells that are censored in the fitting of individual models ranged from 0–43% of the study area. This may, at first, seem troublesome, however, BA ranged from 0.997–1 for all of the individuals indicating that the censored cells possessed very little UD mass and the closed form target and exact censored

UDs were virtually identical. Therefore, the rate censor approximation did not adversely affect parameter estimation in this case.

4 Discussion

This work introduces a continuous-time Markov chain movement process which possesses a closed form log-linear stationary distribution which is a function of spatial covariates. The coefficients in log-linear form allow direct inference of the effect of habitat variables on the utilization distribution. Moreover, because these parameters are estimated within the context of a continuous-time movement model, the temporal correlation between observations is accounted for in the standard errors of the UD coefficients. In addition to UD inference, the fitted movement parameters also provide information on resource selection and local habitat drivers of residency (Hanks et al. 2015; Johnson et al. 2021). Another benefit of the CTMC approach is that barriers to movement can be easily handled in the movement model by simple exclusion of cells on the other side of the barriers. This is especially important for analysis of marine systems such as the FKW example.

In addition to individual inference we also adapted the 2-stage meta-analytic approach of Johnson et al. (2022) to approximate inference from a Bayesian CTMC random effects model. This allows practitioners to estimate population-level space use and movement parameters. Also, information sharing between individuals allows improved inference if movement patterns are similar between individuals. As part of this research, the R package `walk` was developed for processing continuous-space location telemetry data into a discrete-space form and fitting CTMC models using the exact HMM likelihood.

While overarching movement and space-use inference is also shared by Langevin diffusion movement models (Michélot et al. 2019; Dupont et al. 2026), there are benefits to using the CTMC models proposed here. First, although it can be computationally

intensive, one can use the exact likelihood for the movement data. For Langevin diffusion inference an Euler or Ozaki time discretization approximation is needed within a pseudo-likelihood framework for parameter estimation (Gloaguen et al. 2018; Michelot et al. 2019). One might make the argument that the CTMC approach just exchanges discrete space for discrete time approximation. This may be true for constructed variables such as distance from land in the FKW example. However, many habitat variables are only available as cellular rasters, e.g., SST and SLA. Therefore, the Langevin pseudo-likelihoods must also spatially approximate the gradients of these variables as well. Finally, the Langevin models are not well suited to use of categorical habitat characteristics (Michelot et al. 2019) and a smoothing of indicator variables must be used. The CTMC model proposed here can use categorical variables just as easily as continuously valued variables.

Given the apparent positive aspects of the closed-form CTMC approach, it behooves us to look at some potential downsides. The first issue is that CTMC models are built on a defined set of cells, if it is changed, for example, the resolution of a raster grid is increased, it is not apparent how the parameters relate to each other. If the grid is fixed based on habitat covariates, then this is probably not an issue because that is the spatial resolution that habitat information is available. However, if there is some flexibility in resolution of cells then some consideration is warranted. The second issue is that the TPM, $\mathbf{\Gamma}(t)$, can become difficult to compute for very large study areas relative to cell resolution.

Uniformization methods (Sherlock 2021) and rate censoring as implemented in the `walk` package provide ways to mitigate this issue. But for extremely large study areas and very fast moving species (i.e., high $\max(\Lambda_i)$) it can become burdensome. Alternative uniformization methods which calculate likelihood parameter gradients (Rupp et al. 2024) may provide avenues to speed optimization. In addition, Langevin models can possess directional persistence while maintaining the explicit log-linear stationary distributions (Dupont et al. 2026). While CTMC movement models can account for directional

persistence Hanks et al. (2015); Johnson et al. (2021), the stationary distribution loses the log-linear closed form.

Ultimately, bridging the gap between fine-scale movement decisions and long-term space-use remains a central challenge in spatial ecology. By offering a computationally tractable, closed-form bridge between continuous-time movement and utilization distributions, the CTMC framework provides a unified approach to movement and space-use modeling. As telemetry datasets continue to grow in resolution and volume, methods that seamlessly blend animal movement with flexible population-level inference will be essential for management and conservation. The combination of CTMC models for individual movement and meta-analysis methods for population inference provides a theoretical and practical approach for transforming raw telemetry data into scientific inference.

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Conflicts of interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figure 1: Telemetry data for five false killer whales in the Main Hawaiian Island Insular stock and habitat layers for CTMC modeling. The original values are presented here, but the oceanographic variables (B and C) and distance from land (D) were Z -transformed in model fitting.

Figure 2: Parameter estimates for fitted UD surfaces. The Stage 1 values are from individually fitted models. The Stage 2 estimates are derived using the two stage method of Johnson et al. (2022) to approximate inference from fitting a fully joint Bayesian random effects model. The solid vertical lines are the estimated random effect means and the dashed lines are the approximate 95% credible intervals.

Figure 3: Individual Predicted UD's using parameter estimates from the second stage linear mixed model. The parameter estimates, $\bar{\beta}_i$ are approximate maximum *a posteriori* estimates from the posterior $[\beta_i | \mathbf{y}_1, \dots, \mathbf{y}_5]$. The “Typical” UD plot is created from the population mean coefficients.

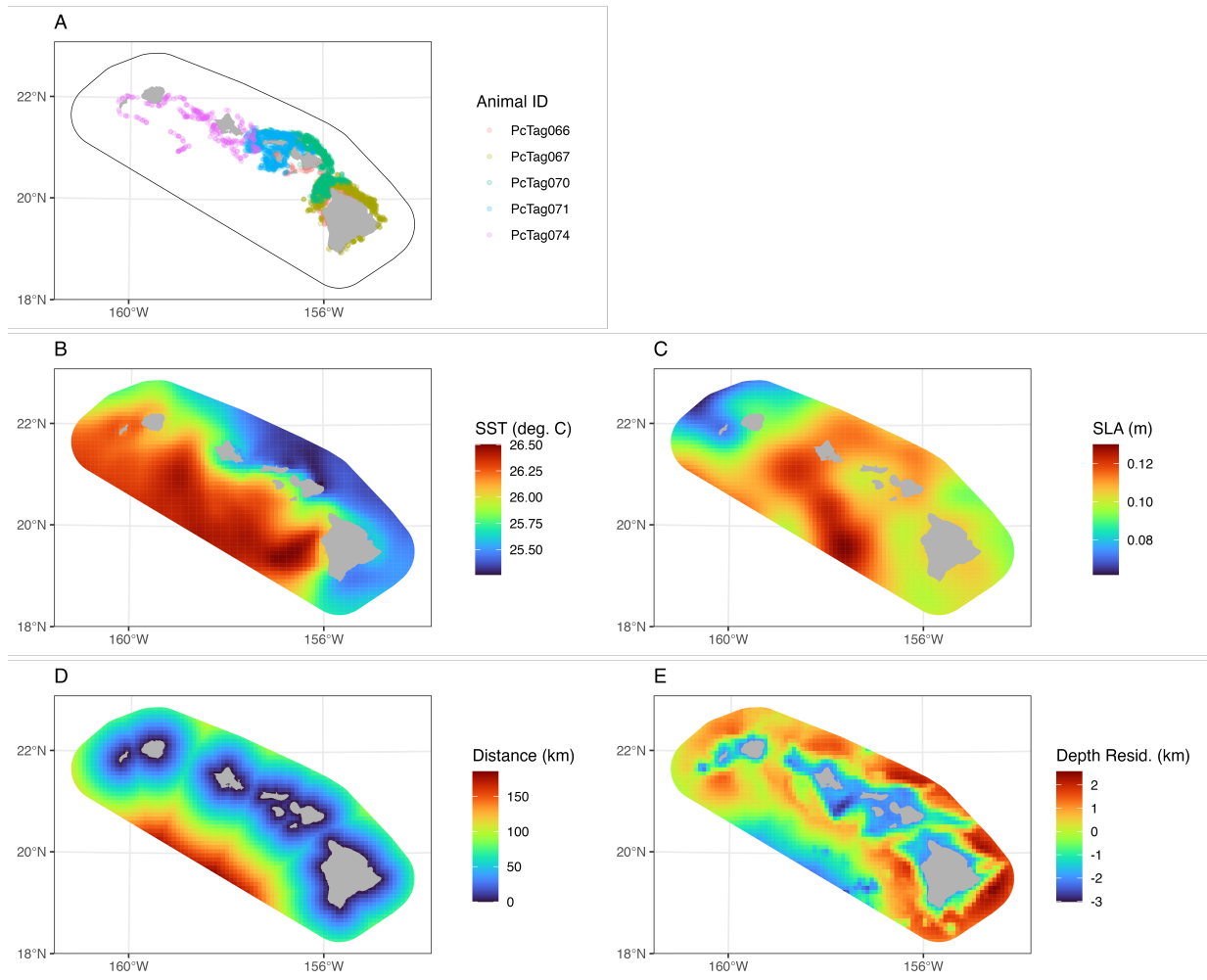


Figure 1

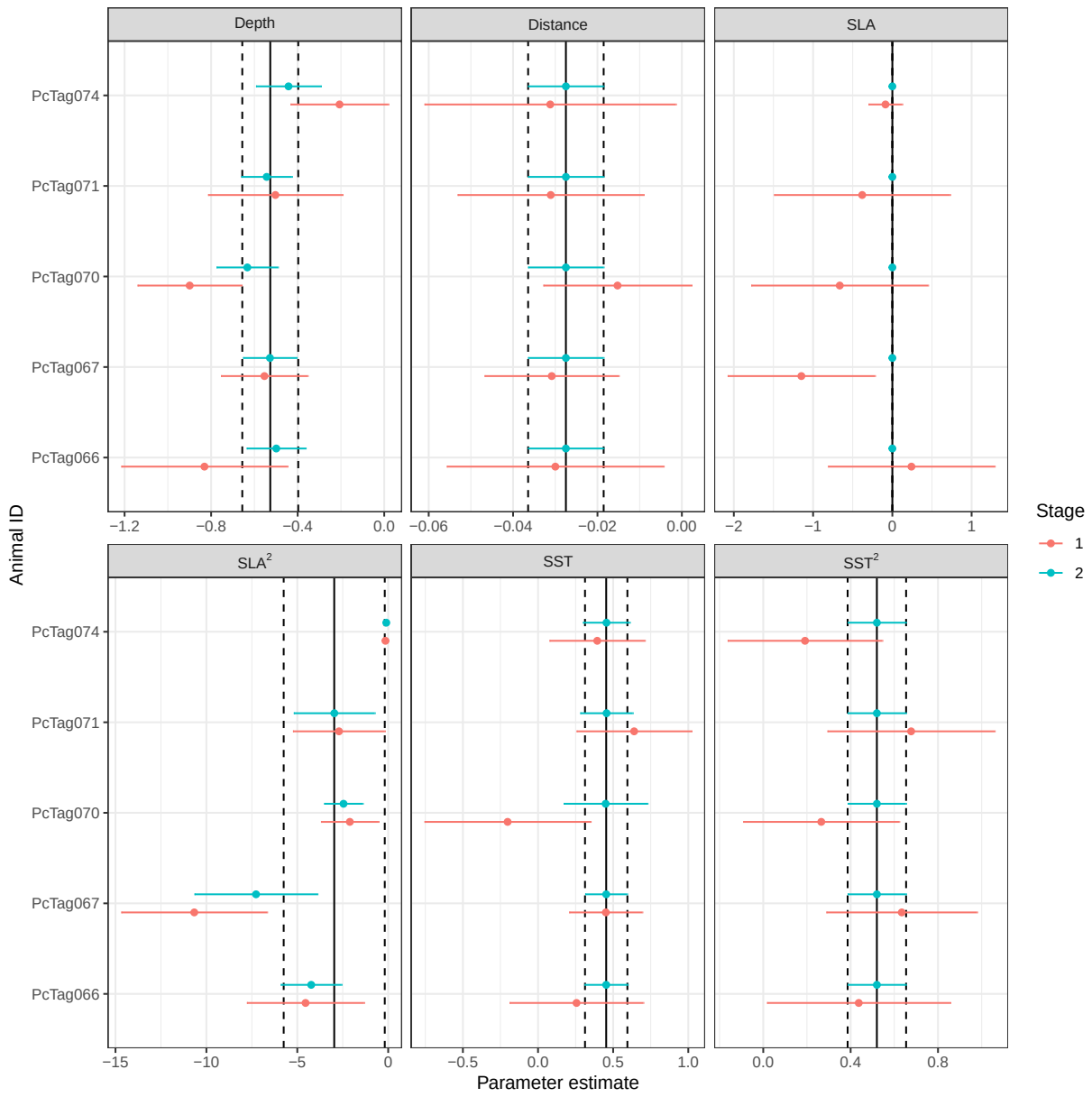


Figure 2

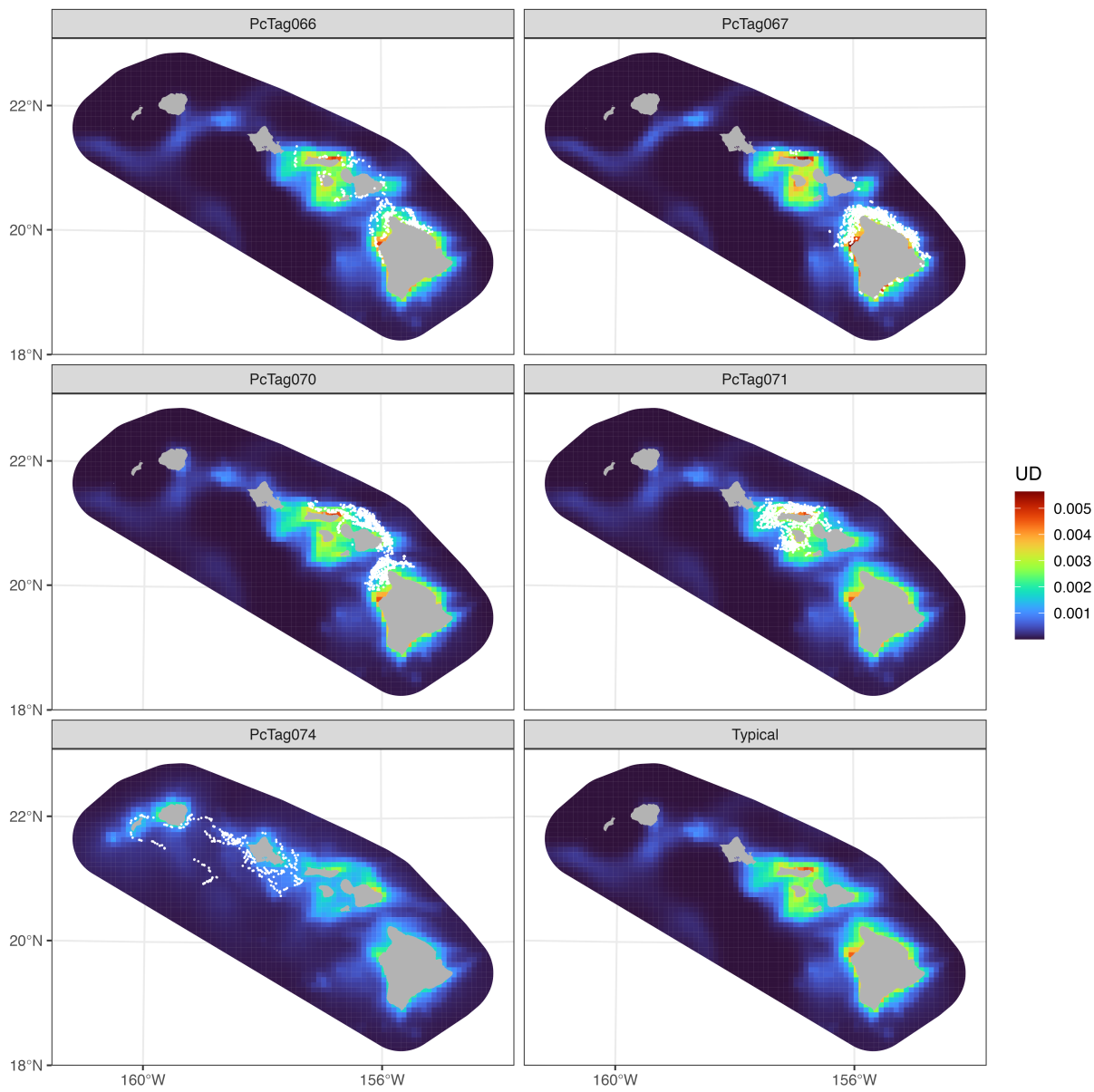


Figure 3